# Which precinct normalizations preserve the linear mixing model?

Notes for `voting-unmixing-ph`

2026-11-09

**Abstract**

Comparing 2013/2016 (row-max normalization) with 2019–2025 (per-ballot normalization) on real returns mixes two effects: the normalization and a large change in the political landscape. This note isolates the first one with a proof and noise-free and noisy toy data. Dividing by the number of ballots keeps the linear mixing model (LMM) exact, with abundances equal to voter fractions. Dividing by the vote sum keeps it exact, but the abundances become vote shares and ballot width is lost. Dividing by the row maximum breaks the sum-to-one LMM whenever the mixed voter types disagree on their top candidate: precincts lie *outside* the simplex of the normalized pure types. No choice of $p$ endmembers fits them exactly. On toy data this inflates MVSA's endmembers and gives them negative loadings that the true types do not have. Figures and the table come from `make_figures.py` (seeded, synthetic, no election data needed). Notation follows Li and Bioucas-Dias (2008), the MVSA paper.

## 1 Setup

**Notation.** Candidates are indexed by $\ell = 1, \ldots, L$, voter types (endmembers) by $i = 1, \ldots, p$, and precincts (pixels) by $j = 1, \ldots, n$. As in the MVSA paper, the LMM is

$$\mathbf{Y} = \mathbf{M}\mathbf{S} \quad \text{s.t.} \quad \mathbf{S} \succeq 0, \;\; \mathbf{1}_p^\top \mathbf{S} = \mathbf{1}_n^\top, \tag{1}$$

with data $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_n]$, mixing matrix $\mathbf{M} = [\mathbf{m}_1, \ldots, \mathbf{m}_p]$ and abundance matrix $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_n]$. Entry $[\mathbf{S}]_{ij}$ is the fraction of endmember $i$ in precinct $j$, so each column $\mathbf{s}_j$ lies in the probability simplex $\Delta_p := \{\mathbf{s} \in \mathbb{R}^p : \mathbf{s} \succeq 0, \; \mathbf{1}_p^\top \mathbf{s} = 1\}$. The MVSA paper writes (1) in the $p$-dimensional signal subspace, where $\mathbf{M}$ is square and $\mathbf{Q} \equiv \mathbf{M}^{-1}$. We stay in $\mathbb{R}^L$ ($\mathbf{M}$ is $L \times p$). The constraints are the same. Estimates carry hats: $\hat{\mathbf{M}}$, $\hat{\mathbf{S}}$.

**Generative model.** Precinct $j$ has $B_j$ ballots. A voter of type $i$ marks candidate $\ell$ with probability $[\mathbf{M}]_{\ell i} \in [0, 1]$, and $[\mathbf{S}]_{ij}$ is the fraction of precinct $j$'s voters of type $i$. Each type has an expected number of names per ballot $c_i := \mathbf{1}_L^\top \mathbf{m}_i$; write $\mathbf{c} = [c_1, \ldots, c_p]^\top$. The expected vote matrix $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_n]$ has columns

$$\underbrace{\mathbf{v}_j}_{\text{vote}} = \underbrace{B_j}_{\text{ballot}} \; \underbrace{\mathbf{M}}_{\text{vote/ballot}} \; \underbrace{\mathbf{s}_j}_{\text{ballot/ballot}} . \tag{2}$$

**Units.** Two things are counted: a *ballot* (one voter, who casts one ballot) and a *vote* (one name marked on one ballot). A *top vote* is a vote for the precinct's top candidate $\ell^* = \arg\max_\ell [\mathbf{v}_j]_\ell$. Units are written under braces or in upright brackets, e.g. [vote/ballot]. A bracket with a subscript, like $[\mathbf{M}]_{\ell i}$, is still a matrix entry. Unit symbols are singular, as in SI ([m/s], [mol/mol]): in [vote/vote]

both sides are the same unit, the vote. Each short unit abbreviates a ratio of two totals over precinct $j$ (or over one type's voters). The finest data are clustered-precinct totals, so [vote/ballot] is a precinct average, never a count on one physical ballot. The denominator is always an observed count: all ballots cast $B_j$, all votes cast $\mathbf{1}_L^\top \mathbf{v}_j$ (marks summed over every candidate and ballot), or the top candidate's votes. Numerators split by type are latent. Spelled out:

| unit | spelled out | where |
|---|---|---|
| vote/ballot | votes for $\ell$ / all ballots cast in $j$ | per-ballot $\mathbf{y}_j$ |
| ballot/ballot | type-$i$ ballots / all ballots cast in $j$ (voter fraction) | per-ballot abundance |
| vote/vote | votes for $\ell$ / all votes cast in $j$ | vote-sum $\mathbf{y}_j$ |
| vote/vote | type-$i$ votes / all votes cast in $j$ (vote share) | vote-sum abundance |
| vote/top vote | votes for $\ell$ / votes for $\ell^*$ in $j$ | row-max $\mathbf{y}_j$ |

For a type's profile the denominator is that type's ballots: $[\mathbf{M}]_{\ell i}$ is [type-$i$ votes for $\ell$ / type-$i$ ballots] and $c_i$ is [type-$i$ votes / type-$i$ ballots]. Same-kind ratios are dimensionless; the unit records which whole they are a fraction of, which matters (see *Units check*).

| symbol | meaning | units |
|---|---|---|
| $B_j$ | ballots cast in precinct $j$ | ballot |
| $[\mathbf{v}_j]_\ell$ | votes for candidate $\ell$ in precinct $j$ | vote |
| $[\mathbf{M}]_{\ell i}$ | chance a type-$i$ ballot marks $\ell$ | type-$i$ votes for $\ell$ / type-$i$ ballots |
| $[\mathbf{S}]_{ij}$ | voter fraction of type $i$ in precinct $j$ | type-$i$ ballots / all ballots in $j$ |
| $c_i$ | names per type-$i$ ballot | type-$i$ votes / type-$i$ ballots |

Sampling noise is set aside until Section 3. Everything below holds for the expectation, i.e. in the limit of large $B_j$.

The abundance sum-to-one constraint (ASC) $\mathbf{1}_p^\top \mathbf{S} = \mathbf{1}_n^\top$ is what matters below. MVSA imposes it ($\mathbf{1}_p^\top \mathbf{Q}\mathbf{Y} = \mathbf{1}_n^\top$), and so does SUNSAL with `addone="yes"`. If the ASC is dropped, $\mathbf{Y} = \mathbf{M}\mathbf{S}$ with $\mathbf{S} \succeq 0$ alone is a cone model.

**Normalizations.** Three are in use or under consideration:

$$\begin{aligned} \text{per-ballot:} &\quad \mathbf{y}_j = \mathbf{v}_j / B_j && \text{[vote/ballot]}, \\ \text{vote-sum:} &\quad \mathbf{y}_j = \mathbf{v}_j / \mathbf{1}_L^\top \mathbf{v}_j && \text{[vote/vote]}, \\ \text{row-max:} &\quad \mathbf{y}_j = \mathbf{v}_j / \max_\ell [\mathbf{v}_j]_\ell && \text{[vote/top vote]}. \end{aligned}$$

The last two have the form $\mathbf{y} = \mathbf{v}/d(\mathbf{v})$, with $d$ positively homogeneous of degree one, $d(\alpha\mathbf{v}) = \alpha\, d(\mathbf{v})$ for $\alpha > 0$. Then $B_j$ cancels: $\mathbf{y}_j = \mathbf{M}\mathbf{s}_j / d(\mathbf{M}\mathbf{s}_j)$.

In the single-precinct statements below we drop the index $j$: $\mathbf{s} \in \Delta_p$ is one column of $\mathbf{S}$, with entries $s_i := [\mathbf{S}]_{ij}$.

Table 1 puts the units of every quantity side by side.

| | $\mathbf{y}_j$ | normalized type $\tilde{\mathbf{m}}_i$ | abundance = share of … | width $\mathbf{1}_L^\top \mathbf{y}_j$ | $\Lambda$ |
|---|---|---|---|---|---|
| per-ballot | vote/ballot | $\mathbf{m}_i$ [vote/ballot] | ballots (voter fraction) | names per ballot [vote/ballot] | — |
| vote-sum | vote/vote | $\mathbf{m}_i / c_i$ [vote/vote] | votes (vote share) | 1 [vote/vote] | 1 |
| row-max | vote/top vote | $\mathbf{m}_i / \max_\ell [\mathbf{M}]_{\ell i}$ [vote/own-top vote] | none: weights sum to $\Lambda$ | vote/top vote | $\geq 1$ |

Table 1: Units of each quantity under each normalization.

**Units check.** In ordinary dimensional analysis the vote-sum and row-max ratios are both dimensionless (vote/vote). What the denominator counts still matters, because the LMM is a weighted average: $\mathbf{y}_j = \sum_i [\mathbf{S}]_{ij}\mathbf{m}_i$ only adds up if each abundance is a share of the denominator's quantity, and that quantity is split exactly among the types. Every ballot belongs to exactly one type, so ballot shares sum to one. Every vote is cast by exactly one type, so vote shares sum to one. Top votes are not split this way. Each type's profile is divided by its *own* top candidate $\ell(i)$, while the precinct is divided by the precinct's top candidate $\ell^*$. Proposition 5 makes this exact.

## 2 Results

**Proposition 1** (per-ballot)**.**

$$\underbrace{\mathbf{y}_j}_{\textit{vote/ballot}} = \underbrace{\mathbf{M}}_{\textit{vote/ballot}}\ \underbrace{\mathbf{s}_j}_{\textit{ballot/ballot}}, \qquad \underbrace{\mathbf{1}_L^\top \mathbf{y}_j}_{\textit{vote/ballot}} = \underbrace{\mathbf{c}^\top}_{\textit{vote/ballot}}\ \underbrace{\mathbf{s}_j}_{\textit{ballot/ballot}},$$

*i.e.* $\mathbf{Y} = \mathbf{MS}$. *The LMM* (1) *holds exactly with the true mixing matrix and with abundances equal to the voter fractions. The width is the mean number of names per ballot.*

*Proof.* Divide (2) by $B_j$. □

*Remark* 2 (blank and overvoted ballots). A ballot that leaves the senate race blank, or overvotes it (all its senate marks are voided), adds nothing to $\mathbf{v}_j$ but is counted in $B_j$. This is a type with $\mathbf{m}_0 = \mathbf{0}$, so Proposition 1 still holds with $p+1$ types, one at the origin. Such ballots lower the width. If instead the denominator excludes them, $\mathbf{y}_j = \sum_{i\geq 1} \frac{s_i}{1-s_0}\mathbf{m}_i$, which is still an LMM, over voters who cast a counted senate ballot. What matters is that the denominator *counts voters* and does not depend on how they voted. A denominator $r_j B_j$ with a precinct-varying factor $r_j$ [counted ballots / ballots cast] gives $\mathbf{1}_p^\top \hat{\mathbf{s}}_j = 1/r_j$, which violates the ASC: the abundances are then shares of a quantity that the types do not split exactly. The 2025 `numberOfValidBallot` column is such a case: $r_j$ has median 1.08 and differs by region.

**Lemma 3** (homogeneous normalizations)**.** *Let $d$ be positively homogeneous with $d(\mathbf{m}_i) > 0$. Write $\tilde{\mathbf{m}}_i := \mathbf{m}_i/d(\mathbf{m}_i)$ for the normalized pure types and $\tilde{\mathbf{M}} = [\tilde{\mathbf{m}}_1, \ldots, \tilde{\mathbf{m}}_p]$. Then*

$$\mathbf{y} = \sum_{i=1}^{p} w_i(\mathbf{s})\,\tilde{\mathbf{m}}_i, \qquad w_i(\mathbf{s}) = \frac{s_i\, d(\mathbf{m}_i)}{d(\mathbf{Ms})} \geq 0, \qquad \sum_i w_i(\mathbf{s}) = \Lambda(\mathbf{s}) := \frac{\sum_i s_i\, d(\mathbf{m}_i)}{d(\sum_i s_i \mathbf{m}_i)}. \tag{3}$$

*Proof.* $\mathbf{y} = \mathbf{Ms}/d(\mathbf{Ms}) = \sum_i s_i \mathbf{m}_i / d(\mathbf{Ms}) = \sum_i \frac{s_i d(\mathbf{m}_i)}{d(\mathbf{Ms})}\,\tilde{\mathbf{m}}_i$. □

Units: $s_i$ is [ballot/ballot], and $d(\mathbf{m}_i)$ and $d(\mathbf{Ms})$ are both "$d$-quantity per ballot". So $w_i$ is the fraction of the precinct's $d$-quantity that type $i$ contributes, *where type $i$'s contribution is measured with its own $d$.*

So $\mathbf{y}$ lies in the cone of the $\tilde{\mathbf{m}}_i$ in every case. It lies in their *simplex* (the LMM (1) with mixing matrix $\tilde{\mathbf{M}}$) exactly when $\Lambda(\mathbf{s}) = 1$.

**Proposition 4** (vote-sum)**.** *For $d(\mathbf{v}) = \mathbf{1}_L^\top \mathbf{v}$, $\Lambda \equiv 1$. The LMM holds exactly with mixing matrix $\tilde{\mathbf{M}}$, $\tilde{\mathbf{m}}_i = \mathbf{m}_i/c_i$, and abundances*

$$\underbrace{[\hat{\mathbf{S}}]_{ij}}_{\textit{vote/vote}} = \frac{\overbrace{c_i}^{\textit{vote/ballot}}\ \overbrace{[\mathbf{S}]_{ij}}^{\textit{ballot/ballot}}}{\underbrace{\mathbf{c}^\top \mathbf{s}_j}_{\textit{vote/ballot}}}, \qquad \underbrace{\tilde{\mathbf{m}}_i}_{\textit{vote/vote}} = \frac{\overbrace{\mathbf{m}_i}^{\textit{vote/ballot}}}{\underbrace{c_i}_{\textit{vote/ballot}}},$$

*the share of precinct $j$'s* votes *cast by type $i$. The map $\mathbf{s}_j \mapsto \hat{\mathbf{s}}_j$ is a bijection of $\Delta_p$ (inverse $[\mathbf{S}]_{ij} \propto [\hat{\mathbf{S}}]_{ij}/c_i$), but it depends on $\mathbf{c}$, which cannot be identified from the data: every $\mathbf{y}_j$ and every $\tilde{\mathbf{m}}_i$ sums to one. Voter fractions and ballot width are therefore both lost.*

*Proof.* $d$ is linear, so $d(\sum_i s_i \mathbf{m}_i) = \sum_i s_i d(\mathbf{m}_i)$ and $\Lambda = 1$ in (3). Rescaling any $\mathbf{m}_i$ leaves $\tilde{\mathbf{m}}_i$, and hence all data, unchanged, so $\mathbf{c}$ is not identifiable. □

**Proposition 5** (row-max lies outside the simplex)**.** *Let $d(\mathbf{v}) = \max_\ell [\mathbf{v}]_\ell$ and write $\tilde{\mathbf{M}}$ for the row-max-normalized types. Then $\Lambda(\mathbf{s}) \geq 1$, so*

$$\mathbf{y} = \Lambda(\mathbf{s})\, \bar{\mathbf{y}}, \qquad \bar{\mathbf{y}} := \sum_i \frac{w_i(\mathbf{s})}{\Lambda(\mathbf{s})} \tilde{\mathbf{m}}_i \in \operatorname{conv}(\tilde{\mathbf{m}}_1, \ldots, \tilde{\mathbf{m}}_p).$$

*The precinct sits on the ray through a point of the simplex of the normalized pure types, a factor $\Lambda \geq 1$ beyond it. Equality $\Lambda(\mathbf{s}) = 1$ holds iff some candidate $\ell$ is a top candidate of every type present in the precinct, i.e. $[\mathbf{M}]_{\ell i} = \max_{\ell'} [\mathbf{M}]_{\ell' i}$ for all $i$ with $s_i > 0$.*

*Proof.* Let $\ell^*$ be a maximizing index of $\mathbf{Ms}$. Then

$$d(\mathbf{Ms}) = \sum_i s_i [\mathbf{M}]_{\ell^* i} \leq \sum_i s_i \max_{\ell'} [\mathbf{M}]_{\ell' i} = \sum_i s_i\, d(\mathbf{m}_i),$$

so $\Lambda \geq 1$. Since $s_i \geq 0$, equality holds iff $[\mathbf{M}]_{\ell^* i} = \max_{\ell'} [\mathbf{M}]_{\ell' i}$ for every $i$ with $s_i > 0$. If some $\ell$ tops every type present, then $[\mathbf{Ms}]_\ell = \sum_i s_i d(\mathbf{m}_i)$, which is the largest any entry of $\mathbf{Ms}$ can be. So $\ell$ is a maximizer and equality holds. This gives the "iff". □

**Units reading of the proof.** Multiply the numerator and denominator of $w_i$ by $B_j$:

$$w_i = \frac{\overbrace{s_i B_j}^{\text{type-}i\text{ ballots}} \quad \overbrace{d(\mathbf{m}_i)}^{\text{type-}i\text{ votes for }\ell(i)\text{ / type-}i\text{ ballots}}}{\underbrace{B_j\, d(\mathbf{Ms})}_{\text{votes for }\ell^*}} = \frac{\text{type-}i\text{ votes for its own top candidate }\ell(i)}{\text{all votes for the precinct's top candidate }\ell^*},$$

$$\Lambda = \sum_i w_i = \frac{\sum_i (\text{type-}i\text{ votes for }\ell(i))}{\sum_i (\text{type-}i\text{ votes for }\ell^*)} \;\geq\; 1,$$

because each type votes for its own top candidate at least as often as for $\ell^*$. The units match (vote/vote), but the numerator counts votes for different candidates for different types. For vote-sum the numerator would be type-$i$ votes for *anyone*, which do partition the denominator, so $\Lambda = 1$.

*Remark* 6 (row-max as a stand-in for the voter count). Row-max is tempting when there is no ballot count, because its denominator grows with turnout:

$$\underbrace{\max_\ell [\mathbf{v}_j]_\ell}_{\text{top vote}} = \underbrace{B_j}_{\text{ballot}} \quad \underbrace{f_j}_{\text{top vote/ballot}}, \qquad f_j := \max_\ell [\mathbf{Ms}_j]_\ell,$$

where $f_j$ is the top candidate's share of ballots. Replacing $B_j$ by $B_j f_j$ works only in two steps, each with its own condition.

1. *Exact LMM: one candidate $\ell_0$ tops every voter type*, and therefore every precinct. This is the $\Lambda = 1$ case of Proposition 5. The denominator is then $[\mathbf{v}_j]_{\ell_0} = \mathbf{e}_{\ell_0}^\top \mathbf{v}_j$, a *linear* function of the votes. The proof of Proposition 4 with $d = \mathbf{e}_{\ell_0}^\top$ gives an exact LMM with abundances

$$\underbrace{[\hat{\mathbf{S}}]_{ij}}_{\text{top vote/top vote}} = \frac{[\mathbf{S}]_{ij}\,[\mathbf{M}]_{\ell_0 i}}{[\mathbf{M}\mathbf{s}_j]_{\ell_0}},$$

   type $i$'s share of $\ell_0$'s votes, not of the voters.

2. *Voter fractions: $f_j$ is also the same in every precinct.* This needs every type to mark $\ell_0$ at the same rate, $[\mathbf{M}]_{\ell_0 i} = f$ for all $i$, i.e. a candidate on essentially every ballot. Then the denominator is $fB_j$, and Proposition 1 holds up to the constant $f$.

Agreement within each precinct is not enough. If the top candidate differs between precincts, each precinct is divided by a different linear function, and the normalized types $\tilde{\mathbf{m}}_i$ change from precinct to precinct.

In the returns, neither condition holds (Table 2; computed in notebook 13, "Is row-max a stand-in for the voter count?"). Without a ballot count, a denominator that is linear in the votes and the same for every precinct, such as vote-sum (Proposition 4), needs neither condition. The cost is that abundances become vote shares rather than voter fractions.

| | precincts topped by national #1 | different top candidates | $f_j$, median (IQR) | corr(row-max denom., voters) |
|---|---|---|---|---|
| 2013 | 46% (Poe) | 18 | — | — |
| 2016 | 13% (Villanueva); Sotto tops 20% | 26 | — | — |
| 2019 | 42% (Villar) | 30 | 0.60 (0.53–0.66) | 0.83 |
| 2022 | 42% (Padilla) | 32 | 0.54 (0.48–0.60) | 0.76 |
| 2025 | 40% (Go); Aquino tops 35% | 33 | 0.54 (0.46–0.64) | 0.75 |

Table 2: Is row-max a stand-in for the voter count? Senatorial returns at the sweep's preprocessing settings. 2013 and 2016 have no voter count, so $f_j$ and the correlation cannot be computed.

Proposition 5 shows that the *true* types do not give a simplex fit. The next result rules out *any* $p$ endmembers. This matters because MVSA is free to choose its own.

**Theorem 7** (row-max admits no exact $p$-endmember LMM)**.** *Assume (i) $\mathbf{M}$ has full column rank $p \geq 2$; (ii) each type has a unique top candidate $\ell(i) = \arg\max_\ell [\mathbf{M}]_{\ell i}$; and (iii) at least two types differ in their top candidate, $\ell(i_1) \neq \ell(i_2)$. Let $\mathcal{Y} = \{\mathbf{M}\mathbf{s}/\max_\ell [\mathbf{M}\mathbf{s}]_\ell : \mathbf{s} \in \Delta_p\}$ be the set of row-max-normalized precincts. Then $\mathcal{Y}$ is not contained in any affine subspace of dimension $p-1$. Hence there are no $\hat{\mathbf{M}} \in \mathbb{R}^{L\times p}$ and $\hat{\mathbf{s}}(\mathbf{s}) \in \Delta_p$ with $\mathbf{y}(\mathbf{s}) = \hat{\mathbf{M}}\hat{\mathbf{s}}(\mathbf{s})$ for all $\mathbf{s}$. The same holds for any set of precincts whose columns $\mathbf{s}_j$ are dense in an open subset of $\Delta_p$ that meets both regions $R_{\ell(i_1)}$ and $R_{\ell(i_2)}$ defined below.*

*Proof.* For a candidate $\ell$, let $R_\ell = \{\mathbf{s} \in \operatorname{relint}\Delta_p : [\mathbf{M}\mathbf{s}]_\ell > [\mathbf{M}\mathbf{s}]_{\ell'}\ \forall \ell' \neq \ell\}$. It is relatively open, by continuity. By (ii), the vertex $\mathbf{e}_{i_1}$ has the strict maximizer $\ell(i_1)$, so points of $\operatorname{relint}\Delta_p$ near $\mathbf{e}_{i_1}$ lie in $R_{\ell(i_1)}$, which is therefore nonempty. Likewise $R_{\ell(i_2)} \neq \emptyset$.

On $R_\ell$, $\mathbf{y}(\mathbf{s}) = \mathbf{M}\mathbf{s}/(\mathbf{e}_\ell^\top \mathbf{M}\mathbf{s})$ is smooth. Its differential along a tangent direction $\boldsymbol{\delta}$ ($\mathbf{1}_p^\top \boldsymbol{\delta} = 0$) is $D\mathbf{y}\,\boldsymbol{\delta} = (\mathbf{I} - \mathbf{y}\,\mathbf{e}_\ell^\top)\mathbf{M}\boldsymbol{\delta}/(\mathbf{e}_\ell^\top \mathbf{M}\mathbf{s})$. If this vanishes, $\mathbf{M}\boldsymbol{\delta} = \alpha\mathbf{M}\mathbf{s}$ with $\alpha = \mathbf{e}_\ell^\top \mathbf{M}\boldsymbol{\delta}/\mathbf{e}_\ell^\top \mathbf{M}\mathbf{s}$. Then $\boldsymbol{\delta} = \alpha\mathbf{s}$ by (i), and $0 = \mathbf{1}_p^\top \boldsymbol{\delta} = \alpha$, so $\boldsymbol{\delta} = \mathbf{0}$. The differential is injective, so $\mathbf{y}(R_\ell)$ contains a $(p-1)$-dimensional embedded piece and $\dim \operatorname{aff} \mathbf{y}(R_\ell) = p - 1$. Every point of $\mathbf{y}(R_\ell)$ has $y_\ell = 1$, so $\operatorname{aff} \mathbf{y}(R_\ell) \subseteq \{y_\ell = 1\}$.

Write $\ell_1 = \ell(i_1)$ and $\ell_2 = \ell(i_2)$. Suppose $\mathcal{Y} \subseteq \mathcal{A}$ with $\mathcal{A}$ affine and $\dim \mathcal{A} = p - 1$. Then aff $\mathbf{y}(R_{\ell_1}) \subseteq \mathcal{A}$, and the dimensions are equal, so $\mathcal{A} = \text{aff } \mathbf{y}(R_{\ell_1}) \subseteq \{y_{\ell_1} = 1\}$. In the same way $\mathcal{A} \subseteq \{y_{\ell_2} = 1\}$. But for $\mathbf{s} \in R_{\ell_1}$, $y_{\ell_2}(\mathbf{s}) = [\mathbf{Ms}]_{\ell_2}/[\mathbf{Ms}]_{\ell_1} < 1$. This is a contradiction. Finally, any $\{\hat{\mathbf{M}}\hat{\mathbf{s}} : \hat{\mathbf{s}} \in \Delta_p\}$ lies in $\text{aff}(\hat{\mathbf{m}}_1, \ldots, \hat{\mathbf{m}}_p)$, which has dimension at most $p - 1$. The dense-subset version follows because the argument only uses open pieces of $R_{\ell_1}$ and $R_{\ell_2}$. □

**Corollary 8** (what row-max preserves)**.** *Under row-max the cone model is still exact:* $\mathbf{y} = \tilde{\mathbf{M}}\mathbf{w}$ *with* $\mathbf{w} \succeq 0$ *and* $\mathbf{1}_p^\top \mathbf{w} = \Lambda(\mathbf{s}) \geq 1$*. The failure comes from the ASC, which both MVSA and SUNSAL (*`addone="yes"`*) impose. The cone coefficients are* $w_i \propto s_i\, d(\mathbf{m}_i)$*, so voter fractions could be recovered only if the types' top-candidate rates* $d(\mathbf{m}_i)$ *were known.*

*Remark* 9 (row-max width). Under row-max,

$$\underbrace{\mathbf{1}_L^\top \mathbf{y}_j}_{\text{vote/top vote}} = \frac{\overbrace{\mathbf{c}^\top \mathbf{s}_j}^{\text{vote/ballot}}}{\underbrace{\max_\ell [\mathbf{Ms}_j]_\ell}_{\text{top vote/ballot}}} :$$

names per ballot divided by the top candidate's share of ballots. This quantity varies with both ballot width and how concentrated the vote is on one candidate. $\Lambda(\mathbf{s}) - 1$ is largest where the mixed types disagree about the top candidate. So a "width" archetype under row-max partly encodes disagreement over the top candidate. On the real 2019–2025 data, row-max width correlates with names per ballot at only 0.68, 0.73 and 0.40.

## 3 Toy data

**Two candidates (Figure 1).** With $L = p = 2$ the three normalizations can be drawn directly. Per-ballot data lie on the segment between the two profiles, evenly spaced in $t = [\mathbf{S}]_{2j}$. Vote-sum data lie on the segment between the normalized profiles, on the line $y_1 + y_2 = 1$, but unevenly spaced: the abundance is $[\hat{\mathbf{S}}]_{2j} = c_2 t/(c_1(1-t) + c_2 t)$, not $t$. Row-max data follow the boundary of the unit square through $(1, 1)$. They bulge past the segment by the factor $\Lambda$ (Proposition 5). No single segment, i.e. no 2-endmember LMM, contains them (Theorem 7).

**Twelve candidates, three types.** Type 1 votes a wide slate ($c_1 = 5.6$ names per ballot, top candidate 1). Type 2 is medium ($c_2 = 3.7$, top candidate 9). Type 3 is narrow ($c_3 = 2.25$, top candidate 12). Every other candidate gets probability 0.05. Precincts: $n = 4000$, $\mathbf{s}_j \sim \text{Dirichlet}(0.7, 0.7, 0.7)$, $B_j \sim \text{Unif}\{200, \ldots, 799\}$. Over these precincts $\Lambda$ has median 1.48 and maximum 2.31.

Figure 2 is noise-free and uses the true endmembers $\tilde{\mathbf{M}}$. Fitting abundances with SUNSAL (non-negative, sum-to-one) recovers $\mathbf{S}$ exactly under per-ballot. Under vote-sum it recovers the vote-share reweighting of Proposition 4, again with zero residual. The reweighting depends on all of $\mathbf{s}_j$ through $\mathbf{c}^\top \mathbf{s}_j$, hence the band. Under row-max the residual is not zero: its largest value is 1.48, and it grows with $\Lambda - 1$. Dropping the ASC brings the residual back to zero (Corollary 8).

**Blind MVSA on noisy counts (Figure 3, Table 3).** Counts $[\mathbf{V}]_{\ell j} \sim \text{Binomial}(B_j, [\mathbf{Ms}_j]_\ell)$, then each normalization, then the pipeline's MVSA (`extract_endmembers`, $p = 3$, best of 5 seeds by RMSE) and SUNSAL. The "noise floor" is the RMSE of the cone fit with the true endmembers;

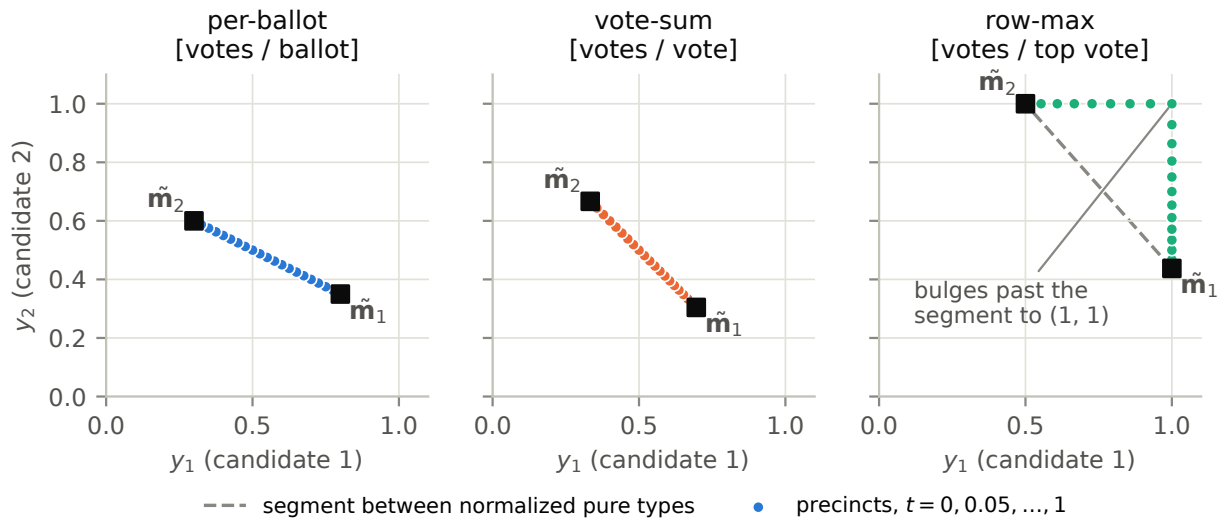



Figure 1: Two candidates, two voter types, noise-free. Type profiles $\mathbf{m}_1 = (0.80, 0.35)$, $\mathbf{m}_2 = (0.30, 0.60)$. Their top candidates differ. Squares mark the normalized pure types. Dots are precincts with type-2 fraction $t$.

it is what sampling noise alone costs. Per-ballot recovers the profiles and their widths, with no negative loadings and RMSE at the floor. Vote-sum also fits at the floor. Its few small negatives sit where the true loadings are near zero ($0.05/c_i$) and come from noise. Row-max is 3× the floor even after MVSA picks its own endmembers. It inflates the column sums by 28–59% and produces negative loadings down to −17% of the largest true loading. The true profiles have no negative loadings, so these come from the normalization.

| | RMSE / noise floor | | | blind MVSA endmembers $\hat{\mathbf{M}}$ | | | |
|---|---|---|---|---|---|---|---|
| normalization | true $\tilde{\mathbf{M}}$ | MVSA | noise floor | min loading | # negative | column sums | true column sums |
| per-ballot | 1.05 | 1.05 | 0.0180 | 1.6% | 0 | 5.76/3.67/2.14 | 5.60/3.70/2.25 |
| vote-sum | 1.00 | 1.00 | 0.0049 | -7.1% | 4 | 1.00/1.00/1.00 | 1.00/1.00/1.00 |
| row-max | 6.02 | 3.03 | 0.0325 | -16.8% | 4 | 10.01/6.52/3.61 | 6.59/4.11/2.81 |

Table 3: Noisy toy election, $p = 3$. RMSE columns are relative to the noise floor (absolute floor in the third column). "Min loading" and "# negative" (loadings below −1%) are relative to the largest true loading, because the scales differ by normalization.

## 4 What this means for the election data

1. Per-ballot normalization is the only one of the three whose abundances are voter fractions and whose endmember column sums are names per ballot (Proposition 1). This requires a denominator that counts voters. For 2025 that means `numberOfActuallyVoters`, not `numberOfValidBallot` (Remark 2).

2. Row-max does not just rescale the per-ballot results. Wherever the mixed groups disagree on the top candidate, it moves precincts off the simplex (Proposition 5), and no set of $p$

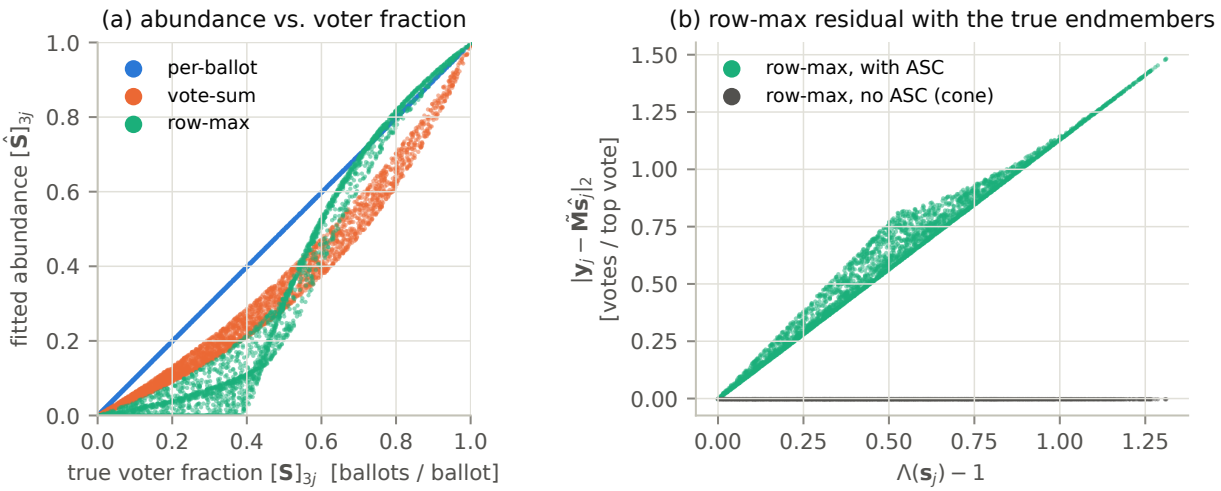



Figure 2: Noise-free toy election, true endmembers. (a) Fitted abundance of the narrow-slate type against its true voter fraction. (b) Row-max residual against $\Lambda(\mathbf{s}_j) - 1$, with and without the sum-to-one constraint. Per-ballot and vote-sum residuals are zero to solver tolerance ($< 10^{-6}$) and are not shown.

endmembers can bring them all back (Theorem 7). On toy data with no negative loadings at all, it produces negative, inflated MVSA endmembers. Some of the 2013/2016 negative loadings may therefore come from the normalization. The 2019–2025 mostly-negative archetype cannot be explained this way, because those years are per-ballot.

3. Switching every year to row-max would make the years comparable in normalization, but by applying this distortion to all of them. Abundances would stop being voter fractions in every year. A cleaner check is to run 2019–2025 under both per-ballot and row-max. Political change is held fixed there, so any difference measures the normalization effect directly.

**Reference.** J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: a fast algorithm to unmix hyperspectral data," in *Proc. IEEE IGARSS*, 2008, vol. 3, pp. III-250–III-253.

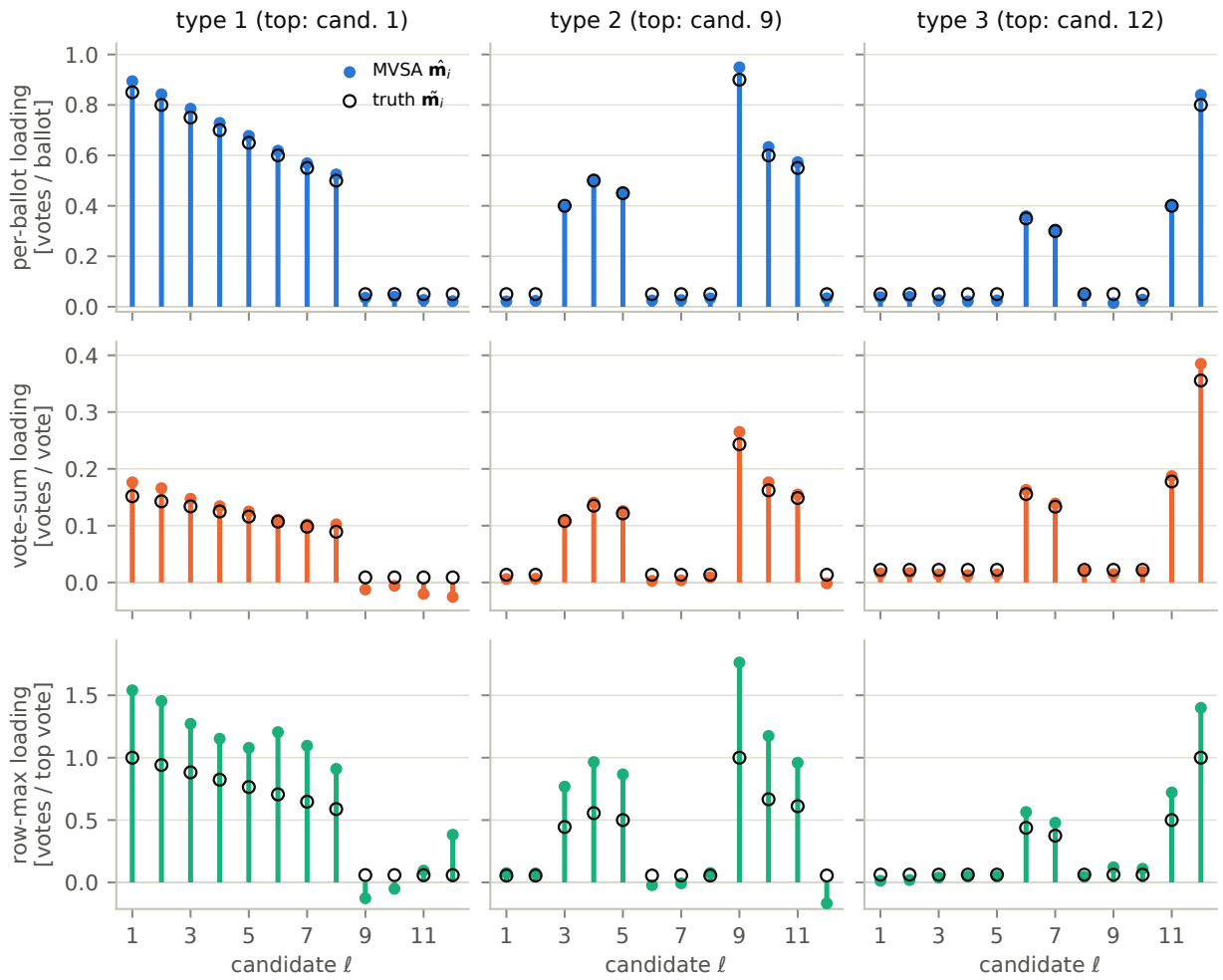



Figure 3: Blind MVSA endmembers $\hat{\mathbf{M}}$ (filled, stems) against the true normalized profiles $\tilde{\mathbf{M}}$ (open circles), matched by cosine. Rows: normalization. Columns: voter type. Each row has its own scale.